

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Classwork 4
  - Due **this Friday** 10/11 @5 pm
  - Solution will be available on Brightspace on 10/12
- Midterm Question Contribution (Bonus Points!)
  - Google Form: https://forms.gle/PQWnzJ3EphU2WyPw8
  - Due this Thursday 10/10 @5 pm
- Grades and Solution of Classwork 3 are posted on Brightspace

Midterm Logistics

- Time: **Tuesday 10/15, 3:40PM to 5:20PM**
- Location: **SEA 205**
- **Conceptual and mathematical**
- No coding questions
- **Closed** book/notes/computer/Internet/classmates/…
- **One** 8.5" x 11" (double-sided) cheat sheet is allowed
- (Optional) You may bring a calculator

Scope

- **Propositional Inference**
  - Lecture 1.2 – 2.2, Classwork 2
- **Probabilistic Inference**
  - Lecture 3.1, Classwork 3
- **Bayesian Networks**
  - Lecture 3.2 – 5.1, Classwork 3 and 4
- **Utility Theory and Decision Networks (excluding VPI)**
  - Lecture 5.2 – 6.1, Classwork 4

Midterm Style

- 5 Questions
  - 4 classwork-style questions
  - 1 question contributed by the class
- “Disclaimers”
  - Midterm Grade on PROWL
  - Midterm Exam

Next…

MIDTERM REVIEW

Overview

KB (Rules + Facts) —inference procedure→ New Facts

1. Propositional Inference

+ uncertainty

2. Probabilistic Inference

Conditioning Rule $P(Q|e) = \frac{P(Q,e)}{P(e)}$

Joint Probability $P(Q, e_1, e_2, \dots)$

+ scalability

Truth Table

World	L	T
w_0	F	F
w_1	F	T
w_2	T	F
w_3	T	T

+ scalability

3. Bayesian Networks

- Structure: DAG
- Semantics: **CPTs**

Symbolic Inference

+ decision

Theorem Proving

1. $R \Rightarrow W$
2. R

3. W

4. Decision Networks

- Utility Theory
- Value of Information

1. Propositional Inference

- ❑ Proposition
- ❑ Logical Sentence
  - ❑ Atomic
  - ❑ Complex
    - ❑ Clause & CNF
- ❑ KB
- ❑ Truth Table
- ❑ Model
  - ❑ World
  - ❑ Model of KB
- ❑ Model Checking
  - ❑ Entailment
    - ❑ Model Enumeration
    - ❑ Theorem Proving
      - ❑ Modus Ponens & Resolution
      - ❑ Proof by Contradiction

Propositional Inference

- **Propositions** (T/F variables) make **assertions** about aspects of the reasoning **environment**.
- Knowledge Base (KB)
  - Rules: general truths
  - Facts: experientially derived
- Query

Propositional Syntax

$$
\begin{aligned}
\textit{Sentence} &\rightarrow \textit{AtomicSentence} \mid \textit{ComplexSentence} \\
\textit{AtomicSentence} &\rightarrow \textit{True} \mid \textit{False} \mid P \mid Q \mid R \mid \ldots \\
\textit{ComplexSentence} &\rightarrow (\ \textit{Sentence}\ ) \mid [\ \textit{Sentence}\ ] \\
&\mid \neg\ \textit{Sentence} \\
&\mid \textit{Sentence} \wedge \textit{Sentence} \\
&\mid \textit{Sentence} \vee \textit{Sentence} \\
&\mid \textit{Sentence} \Rightarrow \textit{Sentence} \\
&\mid \textit{Sentence} \Leftrightarrow \textit{Sentence} \\
\text{OPERATOR PRECEDENCE} &: \neg, \wedge, \vee, \Rightarrow, \Leftrightarrow
\end{aligned}
$$

Figure 7.7 A BNF (Backus–Naur Form) grammar of sentences in propositional logic, along with operator precedences, from highest to lowest.

Propositional Semantics

- A **world** w is a possible instantiation of **all** propositions.
  - One row in a truth table
- **Propositional semantics** define the rules that decide whether a given sentence (e.g., α, β) is **`True`** or **`False`** in some world.

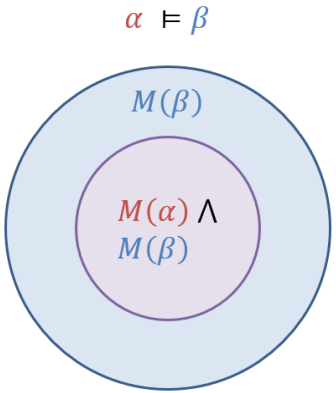
Model

- The **model** of a propositional sentence α is the **set** of all **worlds** $\{w_1, w_2, \dots\}$ in which α is **true**. Formally,

$$M(\alpha) = \{w_1, w_2, \dots\} \text{ where } \alpha \text{ is true}$$

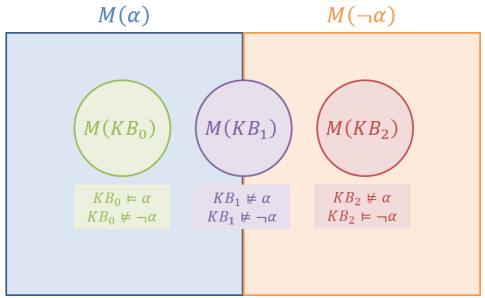
- KB
  - $KB = Rule1 \wedge Rule2 \wedge \cdots \wedge Fact1 \wedge Fact2 \wedge \cdots$
- Model of a KB: $M(KB)$
  - $M(KB \wedge newFact) \subseteq M(KB)$

Model Checking



- Entailment $\models$
  - α **entails** β means β follows from α
  - $\alpha \models \beta$ if the model of α is a **subset** of the model of β
  - $\alpha \models \beta \Leftrightarrow M(\alpha) \subseteq M(\beta)$
- Model Checking
  - Determine if we have **enough information in KB** to make a conclusion
  - Check a model
    - Entailment $\models$: $\mathrm{KB} \models \alpha$ and $\mathrm{KB} \models \neg\alpha$
    - For a query α,

$$KB \models \alpha \Leftrightarrow M(KB) \subseteq M(\alpha)$$



Showing Entailment #1: Model Enumeration

World	R	P	S	$KB = (\neg P \vee S) \wedge \neg R$	$\neg R$	$\neg P$
w_0	F	F	F	T ✓	T ✓	T ✓
w_1	F	F	T	T ✓	T ✓	T ✓
w_2	F	T ✓	F	F	T ✓	F
w_3	F	T ✓	T	T ✓	T ✓	F
w_4	T ✓	F	F	F	F	T ✓
w_5	T ✓	F	T	F	F	T ✓
w_6	T ✓	T ✓	F	F	F	F
w_7	T ✓	T ✓	T	F	F	F

$M(KB)$
$M(R)$
$M(P)$
$M(\neg R)$
$M(\neg P)$

$KB \models R?$ No
$KB \models \neg R?$ Yes
$KB \models P?$ No
$KB \models \neg P?$ No

Showing Entailment #2: Theorem Proving

- Propositional Theorem Proving
  - Inference at the symbolic level
- Rules
  - Rule 1: Modus Ponens $\dfrac{(\alpha \Rightarrow \beta) \wedge \alpha}{\beta}$
  - Rule 2: Resolution

$$\frac{(P_1 \vee P_2 \vee P_i \vee \cdots) \wedge (P_3 \vee P_4 \vee \neg P_i \vee \cdots)}{(P_1 \vee P_2 \vee P_3 \vee P_4 \vee \cdots)}$$

CNF and Resolution

- CNF: a **conjunction** of **clauses**
  - Conjunction: logical **AND** ($\wedge$)
  - Clause: sentence consists exclusively of disjoined (logical **OR** ($\vee$)) propositions ($Clause = P_1 \vee P_2 \vee \cdots \vee P_n$)
- Resolution
  - Applies on a pair of clauses
  - Resolution: an operation defined on a **single** literal at a time.
    - $(\neg S \vee R) \wedge (S \vee \neg R)$
    - $S \vee \neg S \equiv R \vee \neg R \equiv True$

Proof by Contradiction

1. **Assume to the contrary** (i.e., assume $\neg\alpha$ is true)

$$KB = KB \wedge \neg\alpha$$

2. Try to **derive a contradiction**
   - Apply **pairwise resolution** to all clauses, including those that are inferred by resolution **until**:
     - $KB \wedge \neg\alpha \equiv \emptyset$ (reach a contradiction). Therefore, $KB \models \alpha$.
     - There are no other resolutions to consider (do not reach a contradiction). Therefore, $KB \not\models \alpha$.

Logical Equivalence

$$
\begin{aligned}
(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\
(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\
((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\
((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\
\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\
(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\
(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\
\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\
\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\
(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\
(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge
\end{aligned}
$$

Figure 7.11 Standard logical equivalences. The symbols α, β, and γ stand for arbitrary sentences of propositional logic.

2. Probabilistic Inference

- ❑ Probability Theory
  - ❑ Probabilistic Logic Syntax
  - ❑ Properties of Probability Theory
- ❑ Joint and Conditional Distribution
  - ❑ Joint Probability
  - ❑ Conditional Probability
  - ❑ Conditioning Rule
- ❑ Independence

Probabilistic Logic Syntax

Propositional Entity	Probabilistic Analog	Probabilistic Syntax	Interpretation
Proposition	Event / Random Variable	**Capitalized** letters/words	- Need not be binary - Discrete/Continuous
(Truth) Values	(Event) Values	**Lowercase** letters/words	- The value that a random variable attains in a world.
Truth Table	Probability Distribution	$P(A \wedge B \wedge C \wedge \cdots) = P(A, B, C, \ldots)$	- Provides a **functional mapping** between variables and probability values that determines how likely each world is.
Models	Probability Values	$P(A = a, B = b, \ldots) = P(a, b, \ldots)$	- Provides a **single probability value** for any instantiation of variables

Properties of Probability Theory

Rules that distributions must follow:

1. **Probability Value Range**: All probability values assigned to any given world w must range between 0 (impossible) and 1 (certain), inclusively: $0 \leq P(w) \leq 1$
2. **Sum of all probability values** must be 1: $\sum_w P(w) = 1$
3. **Law of Total Probability**: the probability of any subset of variables, α, is the sum of every possible world w that in that subset's models: $\mathrm{P}(\alpha) = \sum_{w \in M(\alpha)} P(w)$

All worlds w in model of α.

Joint and Conditional Distribution

- Joint Probability
  - $P(A, B, C, \dots) = P(A \wedge B \wedge C \wedge \cdots)$
- Conditional Probability
  - $P(Q|E = e) = P(Q|e)$
- Conditioning Rule
  - $P(Q|e) = \frac{P(Q,e)}{P(e)} = \frac{joint\ probability\ of\ \{Q,e\}}{probability\ of\ \{e\}}$
  - $P(Q, e) = P(Q|e) \cdot P(e)$

Independence

- Joint Probability
  - $A \perp\!\!\!\perp B \Leftrightarrow P(A, B) = P(A)P(B)$
- Conditional Probability
  - $A \perp\!\!\!\perp B \Leftrightarrow P(A|B) = P(A)$
- Allows us to factoring large joint probability

3. Bayesian Networks

- ❑ BN
  - ❑ Structure
  - ❑ Semantics
- ❑ Markovian Factorization
- ❑ d-Separation
- ❑ Inference Approaches
  - ❑ Exact Inference
  - ❑ Enumeration Inference

Bayesian Networks (BN)

- Two Components
  - Structure
    - DAG: Directed Acyclic Graph
    - Encode **independence** relationships between variables/nodes
  - Semantics
    - Factor **joint** probability (JPT) using **conditional** probability (CPT) and independence relationships
    - Markovian Factorization:
      - $P(V_1, V_2, \dots) = \prod_{V_i \in V} P(V_i | parents(V_i))$

Probabilistic Inference with BN

- Input
  - **Query**: a query over network variables Q
  - **Evidence**: any observed evidence e
- Output
  - **Conditional distribution:** $P(Q|e)$
- Notation: $BN.ask(Q, e) = P(Q|e)$ Inference Goal
- Also need to determine the **independence** relationships (⫫) between variables from the structure of BN

d-Separation

- 3 Rules

Triplet Type	Format	Rule (Blocked if …)
Fork	$X \leftarrow Z \rightarrow Y$	Blocked if **Z is given**
Chain	$X \leftarrow Z \leftarrow Y$ $X \rightarrow Z \rightarrow Y$	Blocked if **Z is given**
Sink	$X \rightarrow Z \leftarrow Y$	Blocked if **neither Z nor its descendants are given**

Exact Inference

- O(1) Answerable: CPTs / parameters
- O(|V|) Answerable: + Markovian Factorization
- O($|V|^2$) Answerable: + d-separation

Enumeration Inference

- Step 1: Identify Variable Types Q, e, Y
- Steps 2-4: Compute P(Q|e)

$$P(Q|e) = \frac{P(Q,e)}{P(e)}$$ Step 4: Normalization

$$= \frac{1}{P(e)} \cdot P(Q,e)$$

$$= \frac{1}{P(e)} \cdot \sum_{y \in Y} P(Q,e,y)$$

$$= \frac{1}{\sum_{q \in Q} P(Q,e)} \cdot \sum_{y \in Y} P(Q,e,y)$$ Step 2: P(Q,e)

Step 3: P(e)

4. Decision Networks

- ❑ Utility Theory
  - ❑ Utility Function
  - ❑ EU&MEU
- ❑ Decision Networks
  - ❑ 3 Types of Nodes
  - ❑ Finding MEU and d* for a DN

Utility Theory

- **Utility Function** $U(s, a) = u$
  - mapping "state" and "action" to a "utility score"
- **Expected Utility** (EU)
  - $EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$
- **Maximum Expected Utility** (MEU)
  - $MEU(e) = \max_{a \in A} EU(a|e)$
  - $a^* = \arg\max_{a \in A} EU(a|e)$

Decision Networks

- Bayesian Networks + Utility Theory
- Nodes
  - **Chance** Nodes ◯: BN variables w/ CPTs
  - **Decision** Nodes ▢: choices/actions the agent has to make
  - **Utility** Nodes ◇: utility function
- EU for DN
  - $EU(a|e) = \sum_{s \in S} P(s|do(a), e) \cdot U(pa)$

Finding MEU and d* for a DN

- Step 1: Compute $P(S|A,e)$
  - Step 1.1: Identify Variables
  - Step 1.2: Find $P(Q,e|D) = \sum_y P(Q,e,y|D)$
  - Step 1.3: Find $P(e|D) = \sum_q P(Q=q,e|D)$
  - Step 1.4: Find $P(Q|D,e) = \frac{P(Q,e|D)}{P(e|D)}$
- Step 2: Find $EU(a|e) \forall a \in A$
- Step 3: Find $MEU(e)$ and $d^* = argmax_d EU(a|e)$

Next…

Q&A